

Stat 517
Homework # 4

1. Let $Y \sim \text{Uniform}[0, 1]$ and $X_n = Y^n$. Prove X_n converges in probability to zero.
2. Let the random variable Y_n have a distribution that is $b(n, p)$.
 - (a) Prove that Y_n/n converges in probability to p .
 - (b) Prove that $1 - Y_n/n$ converges in probability to $1 - p$.
 - (c) Prove that $(Y_n/n)(1 - Y_n/n)$ converges in probability to $p(1 - p)$.
3. Let W_n denote a random variable with mean μ and variance b/n^p , where $p > 0$, μ and b are constants (not function of n). Prove that W_n converges in probability to μ .
4. Let $X_1, \dots, X_n$ be iid random variables with common pdf

$$f(x) = \begin{cases} e^{-(x-\theta)} & x > \theta, -\infty < \theta < \infty \\ 0 & \text{elsewhere.} \end{cases}$$

This pdf is called the shifted exponential. Let $Y_n = \min \{X_1, \dots, X_n\}$. Prove that $Y_n \rightarrow \theta$ in probability by first obtaining the cdf of Y_n .

5. Let Y_1 denote the minimum of a random sample size n from a distribution that has pdf $f(x) = e^{-(x-\theta)}$, $\theta < x < \infty$, zero elsewhere. Let $Z_n = n(Y_1 - \theta)$. Investigate the limiting distribution of Z_n .
6. Let Y_n denote the maximum of a random sample of size n from a distribution of the continuous type that has cdf $F(x)$ and pdf $f(x) = F'(x)$. Find the limiting distribution of $Z_n = n[1 - F(Y_n)]$.
7. Let Y be $b(72, 1/3)$. Approximate $P(22 \leq Y \leq 28)$.
8. Let $f(x) = 1/x^2$, $1 < x < \infty$, zero elsewhere, be the pdf of a random variable X . Consider a random sample of size 72 from the distribution having this pdf. Compute approximately the probability that more than 50 of the observations of the random sample are less than 3.

1. Let $Y \sim \text{Uniform}[0, 1]$ and $X_n = Y^n$. Prove X_n converges in probability to zero.

• $X_n \xrightarrow{P} 0$ means for every $\epsilon > 0$, $P(|X_n - 0| \geq \epsilon) \rightarrow 0$

• Since $0 \leq Y \leq 1$, $0 \leq Y^n \leq 1$, so $|X_n - 0| = X_n = Y^n$

$$P(|X_n - 0| \geq \epsilon) = P(Y^n \geq \epsilon) = P(Y \geq \epsilon^{1/n}) = P(Y \geq \epsilon^{1/n}) = 1 - \epsilon^{1/n}$$

$n^{\text{th}} \text{ root}$ $Y \sim \text{Uniform}[0, 1]:$
 $P(Y \geq a) = 1 - a \text{ for } 0 < a < 1$

• $\lim_{n \rightarrow \infty} \epsilon^{1/n} = 1$, so $\lim_{n \rightarrow \infty} (1 - \epsilon^{1/n}) = 0$, therefore $P(|X_n - 0| \geq \epsilon) \rightarrow 0$
 $X_n = Y^n \xrightarrow{P} 0$

2. Let the random variable Y_n have a distribution that is $b(n, p)$.

(a) Prove that Y_n/n converges in probability to p .

(b) Prove that $1 - Y_n/n$ converges in probability to $1 - p$.

(c) Prove that $(Y_n/n)(1 - Y_n/n)$ converges in probability to $p(1 - p)$.

$\rightarrow Y_n = X_1 + \dots + X_n$ where each $X_i \sim \text{Bernoulli}(p)$ and $E(X_i) = p$, $\frac{Y_n}{n} = \frac{X_1 + \dots + X_n}{n} = \bar{X}_n$.

a) By the Weak Law of Large Numbers, $X_n \xrightarrow{P} E(X_i)$

so $\frac{Y_n}{n} \xrightarrow{P} p$

iid Bernoulli var., sample mean converges in prob. to p .

b) From part (a), $1 - \frac{Y_n}{n} \xrightarrow{P} 1 - p = 1 - \frac{Y_n}{n} \xrightarrow{P} 1 - p$ subtract both sides from one

so $1 - \frac{Y_n}{n} \xrightarrow{P} 1 - p$ using the continuous mapping theorem

c) From part (a) + (b), $\frac{Y_n}{n} \xrightarrow{P} p$ and $1 - \frac{Y_n}{n} \xrightarrow{P} 1 - p$, $(\frac{Y_n}{n})(1 - \frac{Y_n}{n}) \xrightarrow{P} p(1 - p)$

so $(\frac{Y_n}{n})(1 - \frac{Y_n}{n}) \xrightarrow{P} p(1 - p)$ product for convergence in prob.

using the continuous mapping theorem

3. Let W_n denote a random variable with mean μ and variance b/n^p , where $p > 0$, μ and b are constants (not function of n). Prove that W_n converges in probability to μ . $W_n \xrightarrow{P} \mu$

$$E(W_n) = \mu$$

$$\text{Var}(W_n) = b/n^p, p > 0$$

- By Chebyshev's inequality, $P(|W_n - \mu| \geq \epsilon) \leq \frac{\text{Var}(W_n)}{\epsilon^2}$
- Substitute the variance gives $P(|W_n - \mu| \geq \epsilon) \leq \frac{b}{n^p \epsilon^2}$
- Since $p > 0$, $n^p \rightarrow \infty$, meaning $\frac{b}{n^p \epsilon^2} \rightarrow 0$
- Therefore, $P(|W_n - \mu| \geq \epsilon) \rightarrow 0$ for every $\epsilon > 0$. Hence, $W_n \xrightarrow{P} \mu$.
variance $\downarrow$ as $n \uparrow$. so W_n is $\uparrow$ concentrated around μ .

4. Let $X_1, \dots, X_n$ be iid random variables with common pdf $\text{Convergence in prob.}$

$$f(x) = \begin{cases} e^{-(x-\theta)} & x > \theta, -\infty < \theta < \infty \\ 0 & \text{elsewhere.} \end{cases}$$

This pdf is called the shifted exponential. Let $Y_n = \min\{X_1, \dots, X_n\}$. Prove that $Y_n \rightarrow \theta$ in probability by first obtaining the cdf of Y_n .

- Since $Y_n \geq \theta$, $|Y_n - \theta| \geq \epsilon \rightarrow Y_n - \theta \geq \epsilon \rightarrow Y_n \geq \theta + \epsilon$
- So $P(|Y_n - \theta| \geq \epsilon) = P(Y_n \geq \theta + \epsilon)$
- Using survival probability:
 $P(Y_n \geq \theta + \epsilon) = e^{-n\epsilon}$, as $n \rightarrow \infty$, $e^{-n\epsilon} \rightarrow 0$,

Therefore $P(|Y_n - \theta| \geq \epsilon) \rightarrow 0$,
so $Y_n \xrightarrow{P} \theta$

$$X_1 > y, X_2 > y, \dots, X_n > y \text{ independent}$$

$$P(Y_n > y) = [P(X > y)]^n, \text{ for } y > \theta$$



$$\boxed{P(Y_n > y)}: P(X > y) = e^{-(y-\theta)},$$

$$P(Y_n > y) = (e^{-(y-\theta)})^n = e^{-n(y-\theta)}$$

Therefore, $F_{Y_n}(y) = P(Y_n \leq y) = 1 - e^{-n(y-\theta)}$
for $y > \theta$

$$\boxed{x > \theta:}$$

$$F(x) = \int_{\theta}^x e^{-(t-\theta)} dt = 1 - e^{-(x-\theta)}$$

$$1 - F(x) = e^{-(x-\theta)}$$

Limiting Distribution

5. Let Y_1 denote the minimum of a random sample size n from a distribution that has pdf $f(x) = e^{-(x-\theta)}$, $\theta < x < \infty$, zero elsewhere. Let $Z_n = n(Y_1 - \theta)$. Investigate the limiting distribution of Z_n as $n \rightarrow \infty$.

$$Y_1 = \min(X_1, \dots, X_n)$$

- cdf of Y_1**
- From Q4, $F(x) = 1 - e^{-(x-\theta)}$, $x > \theta$. If Y_1 is the minimum, $P(Y_1 > y) = [1 - F(y)]^n$
 - From Q4, $1 - F(y) = e^{-(y-\theta)}$, $P(Y_1 > y) = e^{-n(y-\theta)}$, so $F_{Y_1}(y) = 1 - e^{-n(y-\theta)}$, $y > \theta$.
- cdf of Z_n**
- By definition, $Z_n = n(Y_1 - \theta)$. Distribution: $P(Z_n \leq z) = P(n(Y_1 - \theta) \leq \frac{z}{n}) = P(Y_1 \leq \theta + \frac{z}{n})$
 using the cdf of Y_1 : $P(Y_1 \leq \theta + \frac{z}{n}) = 1 - e^{-n(\theta + \frac{z}{n} - \theta)} = 1 - e^{-z}$
 'n' disappears
 - Therefore, $F_{Z_n}(z) = 1 - e^{-z}$, $z > 0$

- CDF no longer depends on n , $\lim_{n \rightarrow \infty} F_{Z_n}(z) = 1 - e^{-z}$
- Therefore, $Z_n \xrightarrow{D} \text{Exponential}(1)$ cdf of exponential dist. with rate 1.

$$Y_n = \max(X_1, \dots, X_n)$$

6. Let Y_n denote the maximum of a random sample of size n from a distribution of the continuous type that has cdf $F(x)$ and pdf $f(x) = F'(x)$. Find the limiting distribution of $Z_n = n[1 - F(Y_n)]$. Let $M_n = \max$ of n independent Uniform $(0,1)$ var.

- For a continuous dist., $F(X_i) \sim \text{Uniform}(0,1)$. Since Y_n is the largest X_i , $F(Y_n)$ is also the largest of $F(X_1), \dots, F(X_n)$.
- $M_n = F(Y_n)$, for $0 < m < 1$, $P(M_n \leq m) = m^n = P(U_1 \leq m, \dots, U_n \leq m)$
- For $z > 0$, $P(Z_n \leq z) = P(\frac{n(1 - M_n)}{n} \leq \frac{z}{n}) = P(1 - M_n \leq \frac{z}{n}) = P(M_n \geq 1 - \frac{z}{n}) = 1 - P(M_n < 1 - \frac{z}{n})$
 complement
- Since $P(M_n \leq m) = m^n$, $P(Z_n \leq z) = 1 - (1 - \frac{z}{n})^n$
 Rearrange
 $\lim_{n \rightarrow \infty} P(Z_n \leq z) = 1 - e^{-z}$

$$F_Z(z) = \begin{cases} 0, & z < 0 \\ 1 - e^{-z}, & z \geq 0 \end{cases} \text{ exponential dist. with rate 1.}$$

$$Z_n \xrightarrow{D} \text{Exponential}(1)$$

Binomial

7. Let Y be $b(72, 1/3)$. Approximate $P(22 \leq Y \leq 28)$.

- Binomial Distribution: $\mu = np = 72 \left(\frac{1}{3}\right) = 24$

$$\sigma^2 = np(1-p) = 72 \left(\frac{1}{3}\right) \left(\frac{2}{3}\right) = 16$$

$$\sigma = 4$$

- USE continuity correction + standardize:

$$\star P(22 \leq Y \leq 28) \approx P(21.5 < Y < 28.5)$$

$$P\left(\frac{21.5-24}{4} < Z < \frac{28.5-24}{4}\right) = P(-0.625 < Z < 1.125)$$

- Normal Table: $P(Z < 1.125) - P(Z < -0.625) \approx 0.8697 - 0.2660$

$$P(22 \leq Y \leq 28) \approx 0.604$$

- $P(Y=a)$

$$\rightarrow P(a-0.5 < Y < a+0.5)$$

- $P(Y \leq a)$

$$\rightarrow P(Y < a+0.5)$$

- $P(Y < a)$

$$\rightarrow P(Y < a-0.5)$$

$$\star P(a \leq Y \leq b)$$

$$\rightarrow P(a-0.5 < Y < b+0.5)$$

8. Let $f(x) = 1/x^2, 1 < x < \infty$, zero elsewhere, be the pdf of a random variable X . Consider a random sample of size 72 from the distribution having this pdf. Compute approximately the probability that more than 50 of the observations of the random sample are less than 3.

$$P(X < 3) = \int_1^3 \frac{1}{x^2} dx = \int_1^3 x^{-2} dx = \left[-\frac{1}{x}\right]_1^3 = -\frac{1}{3} - (-1) = \frac{2}{3}, \text{ Hence } Y \sim \text{Bin}\left(72, \frac{2}{3}\right)$$

- USE continuity correction + standardize:

$$\star P(X > 50) = P(Y \geq 51) \approx P(Y > 50.5)$$

$$P\left(Z > \frac{50.5-48}{4}\right) = P(Z > 0.625)$$

$$= 1 - \Phi(0.625) \approx 1 - 0.734$$

$$P(X > 50) \approx 0.266$$

$$\mu = np = 72 \left(\frac{2}{3}\right) = 48$$

$$\sigma^2 = np(1-p) = 72 \left(\frac{2}{3}\right) \left(\frac{1}{3}\right) = 16$$

$$\sigma = 4$$